

KarthEEK Mukkavilli

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Education

University of Texas at Austin

Austin, TX

B.S. Electrical & Computer Engineering — GPA: 3.95/4.0 — Expected May 2028

Coursework: Algorithms · Signals & Systems · Discrete Mathematics · Linear Algebra · Differential Equations · Digital Signal Processing · Circuit Theory

Elements of Computing Certificate (CS) · Business Foundations Minor

Technical Skills

Languages: Python, C++, SQL, TypeScript

Quant/ML: PyTorch, scikit-learn, statsmodels, Kalman Filtering, HMM Regime Detection, Engle-Granger Cointegration, Monte Carlo Simulation, Black-Scholes, Genetic Algorithms

Math: Linear Algebra, Probability & Statistics, Time-Series Analysis, Stochastic Processes

Infra/Tools: Alpaca API, FastAPI, Docker, GitLab CI/CD, SQLAlchemy, Git

Experience

Southwest Research Institute

San Antonio, TX

Student Engineer, Tactical Aerospace Department (Division 16)

May 2026 – Present

- Modernizing a legacy firmware build pipeline in an ITAR-controlled environment, consolidating fragmented codebases into a single validated, version-controlled system.
- Validated build reproducibility across development environments, laying groundwork for full CI/CD automation — rigorous verification methodology applicable to research pipeline validation.
- Building a Dockerized, GitLab CI/CD-automated build and validation pipeline to replace a manual, GUI-driven workflow — direct experience with reproducible, production-grade automation.

Human Centered Robotics Laboratory — UT Austin

Austin, TX

Research Assistant

Jan 2026 – Present

- Trained a PyTorch Transformer on high-frequency glove telemetry to predict 16-DOF bionic hand motion, achieving 1ms sensor-to-actuation latency — directly analogous to low-latency signal-to-execution pipelines.
- Applied PID control and real-time signal processing across 16 joints, improving tracking consistency from unreliable to stable — quantitative systems tuning under real-time constraints.
- Processed and cleaned high-frequency, multi-channel sensor telemetry, building intuition for handling noisy time-series data at scale.

National Oilwell Varco

Houston, TX

Embedded Systems Intern

Jun – Aug 2025

- Architected a 40-node distributed IoT sensor network with reliable data ingestion pipelines to AWS IoT Core — experience applicable to market data collection and distributed systems design.
- Replaced \$20K/site legacy system with \$250 custom board (98% cost reduction); automated deployment tooling cutting onboarding from hours to under one minute across 50 field units.
- Automated TLS certificate provisioning via Python scripting, reducing per-unit onboarding overhead across all 50 deployed units.

Projects

QuantClaw — Algorithmic Trading Engine | Python, Alpaca API, Isolation Forest, Groq API

- Genetic algorithm breeds 50 VWAP+RSI strategy variants over 50 generations daily, selecting the fittest bot for live 5-min bar trading under a T+1 cash-settlement risk manager.
- 12-month backtest: 70% win rate over 250 trades, +20% net return. Live paper trading: +\$160 on \$800 account in one month.
- Isolation Forest anomaly detector monitors real-time volatility; layered in an autonomous Groq LLM engine for discretionary trade decisions under hard-coded risk limits.

Statistical Arbitrage Engine | Python, statsmodels

- Built a pairs-trading engine using Engle-Granger cointegration testing, Kalman filter dynamic hedge ratios, and HMM regime filtering; OLS spread strategy achieved a backtested Sharpe ratio of 1.74.
- Automated periodic cointegration re-screening across candidate pairs, sizing positions off the spread's z-score to manage risk.

Options Pricing & Greeks Dashboard | Python, NumPy

- Implemented Black-Scholes and Monte Carlo pricing engines from scratch, using antithetic variates to reduce Monte Carlo standard error by ~40%; built a live IV surface from 2,931 real-time SPY quotes.
- Cross-validated analytical vs. simulated Greeks (Delta, Gamma, Vega, Theta) across the surface to confirm pricing model consistency.

kinematic-transformer | Python, PyTorch

- Transformer encoder (~69K params) forecasting multi-joint kinematic sequences; 0.003108 validation MSE, 0.571ms inference latency vs. 0.656ms for an LSTM baseline.
- Benchmarked against the LSTM baseline on identical synthetic telemetry to isolate architecture as the source of the latency and accuracy gains.

Activities

UT IEEE RAS · TSA Co-Founder (3× National Finals) · 4× Gold PVSA Medalist · SAMPADA Certified Violinist, ABRSM Grade 5